

Table 3. American National Standard Running and Sliding Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches	Class RC 1			Class RC 2			Class RC 3			Class RC 4		
	Clear- ance ^a	Standard Tolerance Limits		Clearance ^a	Standard Tolerance Limits		Clearance ^a	Standard Tolerance Limits		Clearance ^a	Standard Tolerance Limits	
		Hole H5	Shaft g4		Hole H6	Shaft g5		Hole H7	Shaft f6		Hole H8	Shaft f7
Over To	Values shown below are in thousandths of an inch											
0 – 0.12	0.1	+0.2	–0.1	0.1	+0.25	–0.1	0.3	+0.4	–0.3	0.3	+0.6	–0.3
	0.45	0	–0.25	0.55	0	–0.3	0.95	0	–0.55	1.3	0	–0.7
0.12 – 0.24	0.15	+0.2	–0.15	0.15	+0.3	–0.15	0.4	+0.5	–0.4	0.4	+0.7	–0.4
	0.5	0	–0.3	0.65	0	–0.35	1.12	0	–0.7	1.6	0	–0.9
0.24 – 0.40	0.2	+0.25	–0.2	0.2	+0.4	–0.2	0.5	+0.6	–0.5	0.5	+0.9	–0.5
	0.6	0	–0.35	0.85	0	–0.45	1.5	0	–0.9	2.0	0	–1.1
0.40 – 0.71	0.25	+0.3	–0.25	0.25	+0.4	–0.25	0.6	+0.7	–0.6	0.6	+1.0	–0.6
	0.75	0	–0.45	0.95	0	–0.55	1.7	0	–1.0	2.3	0	–1.3
0.71 – 1.19	0.3	+0.4	–0.3	0.3	+0.5	–0.3	0.8	+0.8	–0.8	0.8	+1.2	–0.8
	0.95	0	–0.55	1.2	0	–0.7	2.1	0	–1.3	2.8	0	–1.6
1.19 – 1.97	0.4	+0.4	–0.4	0.4	+0.6	–0.4	1.0	+1.0	–1.0	1.0	+1.6	–1.0
	1.1	0	–0.7	1.4	0	–0.8	2.6	0	–1.6	3.6	0	–2.0
1.97 – 3.15	0.4	+0.5	–0.4	0.4	+0.7	–0.4	1.2	+1.2	–1.2	1.2	+1.8	–1.2
	1.2	0	–0.7	1.6	0	–0.9	3.1	0	–1.9	4.2	0	–2.4
3.15 – 4.73	0.5	+0.6	–0.5	0.5	+0.9	–0.5	1.4	+1.4	–1.4	1.4	+2.2	–1.4
	1.5	0	–0.9	2.0	0	–1.1	3.7	0	–2.3	5.0	0	–2.8
4.73 – 7.09	0.6	+0.7	–0.6	0.6	+1.0	–0.6	1.6	+1.6	–1.6	1.6	+2.5	–1.6
	1.8	0	–1.1	2.3	0	–1.3	4.2	0	–2.6	5.7	0	–3.2
7.09 – 9.85	0.6	+0.8	–0.6	0.6	+1.2	–0.6	2.0	+1.8	–2.0	2.0	+2.8	–2.0
	2.0	0	–1.2	2.6	0	–1.4	5.0	0	–3.2	6.6	0	–3.8
9.85 – 12.41	0.8	+0.9	–0.8	0.8	+1.2	–0.8	2.5	+2.0	–2.5	2.5	+3.0	–2.5
	2.3	0	–1.4	2.9	0	–1.7	5.7	0	–3.7	7.5	0	–4.5
12.41 – 15.75	1.0	+1.0	–1.0	1.0	+1.4	–1.0	3.0	+2.2	–3.0	3.0	+3.5	–3.0
	2.7	0	–1.7	3.4	0	–2.0	6.6	0	–4.4	8.7	0	–5.2
15.75 – 19.69	1.2	+1.0	–1.2	1.2	+1.6	–1.2	4.0	+2.5	–4.0	4.0	+4.0	–4.0
	3.0	0	–2.0	3.8	0	–2.2	8.1	0	–5.6	10.5	0	–6.5

^aPairs of values shown represent minimum and maximum amounts of clearance resulting from application of standard tolerance limits.

Table 4. American National Standard Running and Sliding Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches	Class RC 5			Class RC 6			Class RC 7			Class RC 8			Class RC 9		
	Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits		Clear- ance ^a	Standard Tolerance Limits	
		Hole H8	Shaft e7		Hole H9	Shaft e8		Hole H9	Shaft d8		Hole H10	Shaft c9		Hole H11	Shaft c11
Over To	Values shown below are in thousandths of an inch														
0 – 0.12	0.6	+0.6	– 0.6	0.6	+1.0	– 0.6	1.0	+1.0	– 1.0	2.5	+1.6	– 2.5	4.0	+2.5	– 4.0
	1.6	0	– 1.0	2.2	0	– 1.2	2.6	0	– 1.6	5.1	0	– 3.5	8.1	0	– 5.6
0.12 – 0.24	0.8	+0.7	– 0.8	0.8	+1.2	– 0.8	1.2	+1.2	– 1.2	2.8	+1.8	– 2.8	4.5	+3.0	– 4.5
	2.0	0	– 1.3	2.7	0	– 1.5	3.1	0	– 1.9	5.8	0	– 4.0	9.0	0	– 6.0
0.24 – 0.40	1.0	+0.9	– 1.0	1.0	+1.4	– 1.0	1.6	+1.4	– 1.6	3.0	+2.2	– 3.0	5.0	+3.5	– 5.0
	2.5	0	– 1.6	3.3	0	– 1.9	3.9	0	– 2.5	6.6	0	– 4.4	10.7	0	– 7.2
0.40 – 0.71	1.2	+1.0	– 1.2	1.2	+1.6	– 1.2	2.0	+1.6	– 2.0	3.5	+2.8	– 3.5	6.0	+4.0	– 6.0
	2.9	0	– 1.9	3.8	0	– 2.2	4.6	0	– 3.0	7.9	0	– 5.1	12.8	0	– 8.8
0.71 – 1.19	1.6	+1.2	– 1.6	1.6	+2.0	– 1.6	2.5	+2.0	– 2.5	4.5	+3.5	– 4.5	7.0	+5.0	– 7.0
	3.6	0	– 2.4	4.8	0	– 2.8	5.7	0	– 3.7	10.0	0	– 6.5	15.5	0	– 10.5
1.19 – 1.97	2.0	+1.6	– 2.0	2.0	+2.5	– 2.0	3.0	+2.5	– 3.0	5.0	+4.0	– 5.0	8.0	+6.0	– 8.0
	4.6	0	– 3.0	6.1	0	– 3.6	7.1	0	– 4.6	11.5	0	– 7.5	18.0	0	– 12.0
1.97 – 3.15	2.5	+1.8	– 2.5	2.5	+3.0	– 2.5	4.0	+3.0	– 4.0	6.0	+4.5	– 6.0	9.0	+7.0	– 9.0
	5.5	0	– 3.7	7.3	0	– 4.3	8.8	0	– 5.8	13.5	0	– 9.0	20.5	0	– 13.5
3.15 – 4.73	3.0	+2.2	– 3.0	3.0	+3.5	– 3.0	5.0	+3.5	– 5.0	7.0	+5.0	– 7.0	10.0	+9.0	– 10.0
	6.6	0	– 4.4	8.7	0	– 5.2	10.7	0	– 7.2	15.5	0	– 10.5	24.0	0	– 15.0
4.73 – 7.09	3.5	+2.5	– 3.5	3.5	+4.0	– 3.5	6.0	+4.0	– 6.0	8.0	+6.0	– 8.0	12.0	+10.0	– 12.0
	7.6	0	– 5.1	10.0	0	– 6.0	12.5	0	– 8.5	18.0	0	– 12.0	28.0	0	– 18.0
7.09 – 9.85	4.0	+2.8	– 4.0	4.0	+4.5	– 4.0	7.0	+4.5	– 7.0	10.0	+7.0	– 10.0	15.0	+12.0	– 15.0
	8.6	0	– 5.8	11.3	0	– 6.8	14.3	0	– 9.8	21.5	0	– 14.5	34.0	0	– 22.0
9.85 – 12.41	5.0	+3.0	– 5.0	5.0	+5.0	– 5.0	8.0	+5.0	– 8.0	12.0	+8.0	– 12.0	18.0	+12.0	– 18.0
	10.0	0	– 7.0	13.0	0	– 8.0	16.0	0	– 11.0	25.0	0	– 17.0	38.0	0	– 26.0
12.41 – 15.75	6.0	+3.5	– 6.0	6.0	+6.0	– 6.0	10.0	+6.0	– 10.0	14.0	+9.0	– 14.0	22.0	+14.0	– 22.0
	11.7	0	– 8.2	15.5	0	– 9.5	19.5	0	– 13.5	29.0	0	– 20.0	45.0	0	– 31.0
15.75 – 19.69	8.0	+4.0	– 8.0	8.0	+6.0	– 8.0	12.0	+6.0	– 12.0	16.0	+10.0	– 16.0	25.0	+16.0	– 25.0
	14.5	0	– 10.5	18.0	0	– 12.0	22.0	0	– 16.0	32.0	0	– 22.0	51.0	0	– 35.0

Tolerance limits given in body of table are added to or subtracted from basic size (as indicated by + or – sign) to obtain maximum and minimum sizes of mating parts.

All data above heavy lines are in accord with ABC agreements. Symbols H5, g4, etc. are hole and shaft designations in ABC system. Limits for sizes above 19.69 inches are also given in the ANSI Standard.

Table 5. American National Standard Clearance Locational Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches Over To	Class LC 1			Class LC 2			Class LC 3			Class LC 4			Class LC 5		
	Standard Tolerance Limits			Standard Tolerance Limits			Standard Tolerance Limits			Standard Tolerance Limits			Standard Tolerance Limits		
	Clear- ance ^a	Hole H6	Shaft h5	Clear- ance ^a	Hole H7	Shaft h6	Clear- ance ^a	Hole H8	Shaft h7	Clear- ance ^a	Hole H10	Shaft h9	Clear- ance ^a	Hole H7	Shaft g6
Values shown below are in thousandths of an inch															
0- 0.12	0 0.45	+0.25 0	0 -0.2	0 0.65	+0.4 0	0 -0.25	0 1	+0.6 0	0 -0.4	0 2.6	+1.6 0	0 -1.0	0.1 0.75	+0.4 0	-0.1 -0.35
0.12- 0.24	0 0.5	+0.3 0	0 -0.2	0 0.8	+0.5 0	0 -0.3	0 1.2	+0.7 0	0 -0.5	0 3.0	+1.8 0	0 -1.2	0.15 0.95	+0.5 0	-0.15 -0.45
0.24- 0.40	0 0.65	+0.4 0	0 -0.25	0 1.0	+0.6 0	0 -0.4	0 1.5	+0.9 0	0 -0.6	0 3.6	+2.2 0	0 -1.4	0.2 1.2	+0.6 0	-0.2 -0.6
0.40- 0.71	0 0.7	+0.4 0	0 -0.3	0 1.1	+0.7 0	0 -0.4	0 1.7	+1.0 0	0 -0.7	0 4.4	+2.8 0	0 -1.6	0.25 1.35	+0.7 0	-0.25 -0.65
0.71- 1.19	0 0.9	+0.5 0	0 -0.4	0 1.3	+0.8 0	0 -0.5	0 2	+1.2 0	0 -0.8	0 5.5	+3.5 0	0 -2.0	0.3 1.6	+0.8 0	-0.3 -0.8
1.19- 1.97	0 1.0	+0.6 0	0 -0.4	0 1.6	+1.0 0	0 -0.6	0 2.6	+1.6 0	0 -1	0 6.5	+4.0 0	0 -2.5	0.4 2.0	+1.0 0	-0.4 -1.0
1.97- 3.15	0 1.2	+0.7 0	0 -0.5	0 1.9	+1.2 0	0 -0.7	0 3	+1.8 0	0 -1.2	0 7.5	+4.5 0	0 -3	0.4 2.3	+1.2 0	-0.4 -1.1
3.15- 4.73	0 1.5	+0.9 0	0 -0.6	0 2.3	+1.4 0	0 -0.9	0 3.6	+2.2 0	0 -1.4	0 8.5	+5.0 0	0 -3.5	0.5 2.8	+1.4 0	-0.5 -1.4
4.73- 7.09	0 1.7	+1.0 0	0 -0.7	0 2.6	+1.6 0	0 -1.0	0 4.1	+2.5 0	0 -1.6	0 10.0	+6.0 0	0 -4	0.6 3.2	+1.6 0	-0.6 -1.6
7.09- 9.85	0 2.0	+1.2 0	0 -0.8	0 3.0	+1.8 0	0 -1.2	0 4.6	+2.8 0	0 -1.8	0 11.5	+7.0 0	0 -4.5	0.6 3.6	+1.8 0	-0.6 -1.8
9.85- 12.41	0 2.1	+1.2 0	0 -0.9	0 3.2	+2.0 0	0 -1.2	0 5	+3.0 0	0 -2.0	0 13.0	+8.0 0	0 -5	0.7 3.9	+2.0 0	-0.7 -1.9
12.41- 15.75	0 2.4	+1.4 0	0 -1.0	0 3.6	+2.2 0	0 -1.4	0 5.7	+3.5 0	0 -2.2	0 15.0	+9.0 0	0 -6	0.7 4.3	+2.2 0	-0.7 -2.1
15.75- 19.69	0 2.6	+1.6 0	0 -1.0	0 4.1	+2.5 0	0 -1.6	0 6.5	+4 0	0 -2.5	0 16.0	+10.0 0	0 -6	0.8 4.9	+2.5 0	-0.8 -2.4

^aPairs of values shown represent minimum and maximum amounts of interference resulting from application of standard tolerance limits.

Table 6. American National Standard Clearance Locational Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches Over To	Class LC 6			Class LC 7			Class LC 8			Class LC 9			Class LC 10			Class LC 11		
	Std. Tolerance Limits			Std. Tolerance Limits			Std. Tolerance Limits			Std. Tolerance Limits			Std. Tolerance Limits			Std. Tolerance Limits		
	Clear- ance ^a	Hole H9	Shaft f8	Clear- ance ^a	Hole H10	Shaft e9	Clear- ance ^a	Hole H10	Shaft d9	Clear- ance ^a	Hole H11	Shaft c10	Clear- ance ^a	Hole H12	Shaft	Clear- ance ^a	Hole H13	Shaft
Values shown below are in thousandths of an inch																		
0- 0.12	0.3 1.9	+1.0 0	-0.3 -0.9	0.6 3.2	+1.6 0	-0.6 -1.6	1.0 2.0	+1.6 0	-1.0 -2.0	2.5 6.6	+2.5 0	-2.5 -4.1	4 12	+4 0	-4 -8	5 17	+6 0	-5 -11
0.12- 0.24	0.4 2.3	+1.2 0	-0.4 -1.1	0.8 3.8	+1.8 0	-0.8 -2.0	1.2 4.2	+1.8 0	-1.2 -2.4	2.8 7.6	+3.0 0	-2.8 -4.6	4.5 14.5	+5 0	-4.5 -9.5	6 20	+7 0	-6 -13
0.24- 0.40	0.5 2.8	+1.4 0	-0.5 -1.4	1.0 4.6	+2.2 0	-1.0 -2.4	1.6 5.2	+2.2 0	-1.6 -3.0	3.0 8.7	+3.5 0	-3.0 -5.2	5 17	+6 0	-5 -11	7 25	+9 0	-7 -16
0.40- 0.71	0.6 3.2	+1.6 0	-0.6 -1.6	1.2 5.6	+2.8 0	-1.2 -2.8	2.0 6.4	+2.8 0	-2.0 -3.6	3.5 10.3	+4.0 0	-3.5 -6.3	6 20	+7 0	-6 -13	8 28	+10 0	-8 -18
0.71- 1.19	0.8 4.0	+2.0 0	-0.8 -2.0	1.6 7.1	+3.5 0	-1.6 -3.6	2.5 8.0	+3.5 0	-2.5 -4.5	4.5 13.0	+5.0 0	-4.5 -8.0	7 23	+8 0	-7 -15	10 34	+12 0	-10 -22
1.19- 1.97	1.0 5.1	+2.5 0	-1.0 -2.6	2.0 8.5	+4.0 0	-2.0 -4.5	3.6 9.5	+4.0 0	-3.0 -5.5	5.0 15.0	+6 0	-5.0 -9.0	8 28	+10 0	-8 -18	12 44	+16 0	-12 -28
1.97- 3.15	1.2 6.0	+3.0 0	-1.0 -3.0	2.5 10.0	+4.5 0	-2.5 -5.5	4.0 11.5	+4.5 0	-4.0 -7.0	6.0 17.5	+7 0	-6.0 -10.5	10 34	+12 0	-10 -22	14 50	+18 0	-14 -32
3.15- 4.73	1.4 7.1	+3.5 0	-1.4 -3.6	3.0 11.5	+5.0 0	-3.0 -6.5	5.0 13.5	+5.0 0	-5.0 -8.5	7 21	+9 0	-7 -12	11 39	+14 0	-11 -25	16 60	+22 0	-16 -38
4.73- 7.09	1.6 8.1	+4.0 0	-1.6 -4.1	3.5 13.5	+6.0 0	-3.5 -7.5	6 16	+6 0	-6 -10	8 24	+10 0	-8 -14	12 44	+16 0	-12 -28	18 68	+25 0	-18 -43
7.09- 9.85	2.0 9.3	+4.5 0	-2.0 -4.8	4.0 15.5	+7.0 0	-4.0 -8.5	7 18.5	+7 0	-7 -11.5	10 29	+12 0	-10 -17	16 52	+18 0	-16 -34	22 78	+28 0	-22 -50
9.85- 12.41	2.2 10.2	+5.0 0	-2.2 -5.2	4.5 17.5	+8.0 0	-4.5 -9.5	7 20	+8 0	-7 -12	12 32	+12 0	-12 -20	20 60	+20 0	-20 -40	28 88	+30 0	-28 -58
12.41- 15.75	2.5 12.0	+6.0 0	-2.5 -6.0	5.0 20.0	+9.0 0	-5 -11	8 23	+9 0	-8 -14	14 37	+14 0	-14 -23	22 66	+22 0	-22 -44	30 100	+35 0	-30 -65
15.75- 19.69	2.8 12.8	+6.0 0	-2.8 -6.8	5.0 21.0	+10.0 0	-5 -11	9 25	+10 0	-9 -15	16 42	+16 0	-16 -26	25 75	+25 0	-25 -50	35 115	+40 0	-35 -75

Table 7. ANSI Standard Transition Locational Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches	Class LT 1			Class LT 2			Class LT 3			Class LT 4			Class LT 5			Class LT 6		
	Fit*	Std. Tolerance Limits		Fit*	Std. Tolerance Limits		Fit*	Std. Tolerance Limits		Fit*	Std. Tolerance Limits		Fit*	Std. Tolerance Limits		Fit*	Std. Tolerance Limits	
		Hole H7	Shaft js6		Hole H8	Shaft js7		Hole H7	Shaft k6		Hole H8	Shaft k7		Hole H7	Shaft n6		Hole H7	Shaft n7
Over To	Values shown below are in thousandths of an inch																	
0 – 0.12	-0.12 +0.52	+0.4 0	+0.12 -0.12	-0.2 +0.8	+0.6 0	+0.2 -0.2							-0.5 +0.15	+0.4 0	+0.5 +0.25	-0.65 +0.15	+0.4 0	+0.65 +0.25
0.12 – 0.24	-0.15 +0.65	+0.5 0	+0.15 -0.15	-0.25 +0.95	+0.7 0	+0.25 -0.25							-0.6 +0.2	+0.5 0	+0.6 +0.3	-0.8 +0.2	+0.5 0	+0.8 +0.3
0.24 – 0.40	-0.2 +0.8	+0.6 0	+0.2 -0.2	-0.3 +1.2	+0.9 0	+0.3 -0.3	-0.5 +0.5	+0.6 0	+0.5 +0.1	-0.7 +0.8	+0.9 0	+0.7 +0.1	-0.8 +0.2	+0.6 0	+0.8 +0.4	-1.0 +0.2	+0.6 0	+1.0 +0.4
0.40 – 0.71	-0.2 +0.9	+0.7 0	+0.2 -0.2	-0.35 +1.35	+1.0 0	+0.35 -0.35	-0.5 +0.6	+0.7 0	+0.5 +0.1	-0.8 +0.9	+1.0 0	+0.8 +0.1	-0.9 +0.2	+0.7 0	+0.9 +0.5	-1.2 +0.2	+0.7 0	+1.2 +0.5
0.71 – 1.19	-0.25 +1.05	+0.8 0	+0.25 -0.25	-0.4 +1.6	+1.2 0	+0.4 -0.4	-0.6 +0.7	+0.8 0	+0.6 +0.1	-0.9 +1.1	+1.2 0	+0.9 +0.1	-1.1 +0.2	+0.8 0	+1.1 +0.6	-1.4 +0.2	+0.8 0	+1.4 +0.6
1.19 – 1.97	-0.3 +1.3	+1.0 0	+0.3 -0.3	-0.5 +2.1	+1.6 0	+0.5 -0.5	-0.7 +0.9	+1.0 0	+0.7 +0.1	-1.1 +1.5	+1.6 0	+1.1 +0.1	-1.3 +0.3	+1.0 0	+1.3 +0.7	-1.7 +0.3	+1.0 0	+1.7 +0.7
1.97 – 3.15	-0.3 +1.5	+1.2 0	+0.3 -0.3	-0.6 +2.4	+1.8 0	+0.6 -0.6	-0.8 +1.1	+1.2 0	+0.8 +0.1	-1.3 +1.7	+1.8 0	+1.3 +0.1	-1.5 +0.4	+1.2 0	+1.5 +0.8	-2.0 +0.4	+1.2 0	+2.0 +0.8
3.15 – 4.73	-0.4 +1.8	+1.4 0	+0.4 -0.4	-0.7 +2.9	+2.2 0	+0.7 -0.7	-1.0 +1.3	+1.4 0	+1.0 +0.1	-1.5 +2.1	+2.2 0	+1.5 +0.1	-1.9 +0.4	+1.4 0	+1.9 +1.0	-2.4 +0.4	+1.4 0	+2.4 +1.0
4.73 – 7.09	-0.5 +2.1	+1.6 0	+0.5 -0.5	-0.8 +3.3	+2.5 0	+0.8 -0.8	-1.1 +1.5	+1.6 0	+1.1 +0.1	-1.7 +2.4	+2.5 0	+1.7 +0.1	-2.2 +0.4	+1.6 0	+2.2 +1.2	-2.8 +0.4	+1.6 0	+2.8 +1.2
7.09 – 9.85	-0.6 +2.4	+1.8 0	+0.6 -0.6	-0.9 +3.7	+2.8 0	+0.9 -0.9	-1.4 +1.6	+1.8 0	+1.4 +0.2	-2.0 +2.6	+2.8 0	+2.0 +0.2	-2.6 +0.4	+1.8 0	+2.6 +1.4	-3.2 +0.4	+1.8 0	+3.2 +1.4
9.85 – 12.41	-0.6 +2.6	+2.0 0	+0.6 -0.6	-1.0 +4.0	+3.0 0	+1.0 -1.0	-1.4 +1.8	+2.0 0	+1.4 +0.2	-2.2 +2.8	+3.0 0	+2.2 +0.2	-2.6 +0.6	+2.0 0	+2.6 +1.4	-3.4 +0.6	+2.0 0	+3.4 +1.4
12.41 – 15.75	-0.7 +2.9	+2.2 0	+0.7 -0.7	-1.0 +4.5	+3.5 0	+1.0 -1.0	-1.6 +2.0	+2.2 0	+1.6 +0.2	-2.4 +3.3	+3.5 0	+2.4 +0.2	-3.0 +0.6	+2.2 0	+3.0 +1.6	-3.8 +0.6	+2.2 0	+3.8 +1.6
15.75 – 19.69	-0.8 +3.3	+2.5 0	+0.8 -0.8	-1.2 +5.2	+4.0 0	+1.2 -1.2	-1.8 +2.3	+2.5 0	+1.8 +0.2	-2.7 +3.8	+4.0 0	+2.7 +0.2	-3.4 +0.7	+2.5 0	+3.4 +1.8	-4.3 +0.7	+2.5 0	+4.3 +1.8

Table 8. ANSI Standard Interference Location Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches Over To	Class LN 1			Class LN 2			Class LN 3		
	Limits of Inter- ference	Standard Limits		Lim- its of Inter- ference	Standard Limits		Limits of Inter- ference	Standard Limits	
		Hole H6	Shaft n5		Hole H7	Shaft p6		Hole H7	Shaft r6
	Values shown below are given in thousandths of an inch								
0– 0.12	0	+0.25	+0.45	0	+0.4	+0.65	0.1	+0.4	+0.75
	0.45	0	+0.25	0.65	0	+0.4	0.75	0	+0.5
0.12– 0.24	0	+0.3	+0.5	0	+0.5	+0.8	0.1	+0.5	+0.9
	0.5	0	+0.3	0.8	0	+0.5	0.9	0	+0.6
0.24– 0.40	0	+0.4	+0.65	0	+0.6	+1.0	0.2	+0.6	+1.2
	0.65	0	+0.4	1.0	0	+0.6	1.2	0	+0.8
0.40– 0.71	0	+0.4	+0.8	0	+0.7	+1.1	0.3	+0.7	+1.4
	0.8	0	+0.4	1.1	0	+0.7	1.4	0	+1.0
0.71– 1.19	0	+0.5	+1.0	0	+0.8	+1.3	0.4	+0.8	+1.7
	1.0	0	+0.5	1.3	0	+0.8	1.7	0	+1.2
1.19– 1.97	0	+0.6	+1.1	0	+1.0	+1.6	0.4	+1.0	+2.0
	1.1	0	+0.6	1.6	0	+1.0	2.0	0	+1.4
1.97– 3.15	0.1	+0.7	+1.3	0.2	+1.2	+2.1	0.4	+1.2	+2.3
	1.3	0	+0.8	2.1	0	+1.4	2.3	0	+1.6
3.15– 4.73	0.1	+0.9	+1.6	0.2	+1.4	+2.5	0.6	+1.4	+2.9
	1.6	0	+1.0	2.5	0	+1.6	2.9	0	+2.0
4.73– 7.09	0.2	+1.0	+1.9	0.2	+1.6	+2.8	0.9	+1.6	+3.5
	1.9	0	+1.2	2.8	0	+1.8	3.5	0	+2.5
7.09– 9.85	0.2	+1.2	+2.2	0.2	+1.8	+3.2	1.2	+1.8	+4.2
	2.2	0	+1.4	3.2	0	+2.0	4.2	0	+3.0
9.85– 12.41	0.2	+1.2	+2.3	0.2	+2.0	+3.4	1.5	+2.0	+4.7
	2.3	0	+1.4	3.4	0	+2.2	4.7	0	+3.5
12.41– 15.75	0.2	+1.4	+2.6	0.3	+2.2	+3.9	2.3	+2.2	+5.9
	2.6	0	+1.6	3.9	0	+2.5	5.9	0	+4.5
15.75– 19.69	0.2	+1.6	+2.8	0.3	+2.5	+4.4	2.5	+2.5	+6.6
	2.8	0	+1.8	4.4	0	+2.8	6.6	0	+5.0

Table 9. ANSI Standard Force and Shrink Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches Over To		Class FN 1			Class FN 2			Class FN 3			Class FN 4			Class FN 5			
		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		
			Hole H6	Shaft		Hole H7	Shaft s6		Hole H7	Shaft t6		Hole H7	Shaft u6		Hole H8	Shaft x7	
Values shown below are in thousandths of an inch																	
0– 0.12		0.05 0.5	+0.25 0	+0.5 +0.3	0.2 0.85	+0.4 0	+0.85 +0.6					0.3 0.95	+0.4 0	+0.95 +0.7	0.3 1.3	+0.6 0	+1.3 +0.9
0.12– 0.24		0.1 0.6	+0.3 0	+0.6 +0.4	0.2 1.0	+0.5 0	+1.0 +0.7					0.4 1.2	+0.5 0	+1.2 +0.9	0.5 1.7	+0.7 0	+1.7 +1.2
0.24– 0.40		0.1 0.75	+0.4 0	+0.75 +0.5	0.4 1.4	+0.6 0	+1.4 +1.0					0.6 1.6	+0.6 0	+1.6 +1.2	0.5 2.0	+0.9 0	+2.0 +1.4
0.40– 0.56		0.1 0.8	+0.4 0	+0.8 +0.5	0.5 1.6	+0.7 0	+1.6 +1.2					0.7 1.8	+0.7 0	+1.8 +1.4	0.6 2.3	+1.0 0	+2.3 +1.6
0.56– 0.71		0.2 0.9	+0.4 0	+0.9 +0.6	0.5 1.6	+0.7 0	+1.6 +1.2					0.7 1.8	+0.7 0	+1.8 +1.4	0.8 2.5	+1.0 0	+2.5 +1.8
0.71– 0.95		0.2 1.1	+0.5 0	+1.1 +0.7	0.6 1.9	+0.8 0	+1.9 +1.4					0.8 2.1	+0.8 0	+2.1 +1.6	1.0 3.0	+1.2 0	+3.0 +2.2
0.95– 1.19		0.3 1.2	+0.5 0	+1.2 +0.8	0.6 1.9	+0.8 0	+1.9 +1.4	0.8 2.1	+0.8 0	+2.1 +1.6	+1.0 2.3	+0.8 0	+2.3 +1.8	1.3 3.3	+1.2 0	+3.3 +2.5	
1.19– 1.58		0.3 1.3	+0.6 0	+1.3 +0.9	0.8 2.4	+1.0 0	+2.4 +1.8	1.0 2.6	+1.0 0	+2.6 +2.0	+1.5 3.1	+1.0 0	+3.1 +2.5	1.4 4.0	+1.6 0	+4.0 +3.0	
1.58– 1.97		0.4 1.4	+0.6 0	+1.4 +1.0	0.8 2.4	+1.0 0	+2.4 +1.8	1.2 2.8	+1.0 0	+2.8 +2.2	1.8 3.4	+1.0 0	+3.4 +2.8	2.4 5.0	+1.6 0	+5.0 +4.0	
1.97– 2.56		0.6 1.8	+0.7 0	+1.8 +1.3	0.8 2.7	+1.2 0	+2.7 +2.0	1.3 3.2	+1.2 0	+3.2 +2.5	2.3 4.2	+1.2 0	+4.2 +3.5	3.2 6.2	+1.8 0	+6.2 +5.0	
2.56– 3.15		0.7 1.9	+0.7 0	+1.9 +1.4	1.0 2.9	+1.2 0	+2.9 +2.2	1.8 3.7	+1.2 0	+3.7 +3.0	2.8 4.7	+1.2 0	+4.7 +4.0	4.2 7.2	+1.8 0	+7.2 +6.0	
3.15– 3.94		0.9 2.4	+0.9 0	+2.4 +1.8	1.4 3.7	+1.4 0	+3.7 +2.8	2.1 4.4	+1.4 0	+4.4 +3.5	3.6 5.9	+1.4 0	+5.9 +5.0	4.8 8.4	+2.2 0	+8.4 +7.0	
3.94– 4.73		1.1 2.6	+0.9 0	+2.6 +2.0	1.6 3.9	+1.4 0	+3.9 +3.0	2.6 4.9	+1.4 0	+4.9 +4.0	4.6 6.9	+1.4 0	+6.9 +6.0	5.8 9.4	+2.2 0	+9.4 +8.0	

Table 9. (Continued) ANSI Standard Force and Shrink Fits ANSI B4.1-1967 (R1987)

Nominal Size Range, Inches		Class FN 1			Class FN 2			Class FN 3			Class FN 4			Class FN 5			
		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		Inter- ference ^a	Standard Tolerance Limits		
			Hole H6	Shaft		Hole H7	Shaft s6		Hole H7	Shaft t6		Hole H7	Shaft u6		Hole H8	Shaft x7	
Over	To	Values shown below are in thousandths of an inch															
4.73– 5.52	1.2	+1.0	+2.9	1.9	+1.6	+4.5	3.4	+1.6	+6.0	5.4	+1.6	+8.0	7.5	+2.5	+11.6		
	2.9	0	+2.2	4.5	0	+3.5	6.0	0	+5.0	8.0	0	+7.0	11.6	0	+10.0		
5.52– 6.30	1.5	+1.0	+3.2	2.4	+1.6	+5.0	3.4	+1.6	+6.0	5.4	+1.6	+8.0	9.5	+2.5	+13.6		
	3.2	0	+2.5	5.0	0	+4.0	6.0	0	+5.0	8.0	0	+7.0	13.6	0	+12.0		
6.30– 7.09	1.8	+1.0	+3.5	2.9	+1.6	+5.5	4.4	+1.6	+7.0	6.4	+1.6	+9.0	9.5	+2.5	+13.6		
	3.5	0	+2.8	5.5	0	+4.5	7.0	0	+6.0	9.0	0	+8.0	13.6	0	+12.0		
7.09– 7.88	1.8	+1.2	+3.8	3.2	+1.8	+6.2	5.2	+1.8	+8.2	7.2	+1.8	+10.2	11.2	+2.8	+15.8		
	3.8	0	+3.0	6.2	0	+5.0	8.2	0	+7.0	10.2	0	+9.0	15.8	0	+14.0		
7.88– 8.86	2.3	+1.2	+4.3	3.2	+1.8	+6.2	5.2	+1.8	+8.2	8.2	+1.8	+11.2	13.2	+2.8	+17.8		
	4.3	0	+3.5	6.2	0	+5.0	8.2	0	+7.0	11.2	0	+10.0	17.8	0	+16.0		
8.86– 9.85	2.3	+1.2	+4.3	4.2	+1.8	+7.2	6.2	+1.8	+9.2	10.2	+1.8	+13.2	13.2	+2.8	+17.8		
	4.3	0	+3.5	7.2	0	+6.0	9.2	0	+8.0	13.2	0	+12.0	17.8	0	+16.0		
9.85– 11.03	2.8	+1.2	+4.9	4.0	+2.0	+7.2	7.0	+2.0	+10.2	10.0	+2.0	+13.2	15.0	+3.0	+20.0		
	4.9	0	+4.0	7.2	0	+6.0	10.2	0	+9.0	13.2	0	+12.0	20.0	0	+18.0		
11.03– 12.41	2.8	+1.2	+4.9	5.0	+2.0	+8.2	7.0	+2.0	+10.2	12.0	+2.0	+15.2	17.0	+3.0	+22.0		
	4.9	0	+4.0	8.2	0	+7.0	10.2	0	+9.0	15.2	0	+14.0	22.0	0	+20.0		
12.41– 13.98	3.1	+1.4	+5.5	5.8	+2.2	+9.4	7.8	+2.2	+11.4	13.8	+2.2	+17.4	18.5	+3.5	+24.2		
	5.5	0	+4.5	9.4	0	+8.0	11.4	0	+10.0	17.4	0	+16.0	24.2	0	+22.0		
13.98– 15.75	3.6	+1.4	+6.1	5.8	+2.2	+9.4	9.8	+2.2	+13.4	15.8	+2.2	+19.4	21.5	+3.5	+27.2		
	6.1	0	+5.0	9.4	0	+8.0	13.4	0	+12.0	19.4	0	+18.0	27.2	0	+25.0		
15.75– 17.72	4.4	+1.6	+7.0	6.5	+2.5	+10.6	9.5	+2.5	+13.6	17.5	+2.5	+21.6	24.0	+4.0	+30.5		
	7.0	0	+6.0	10.6	0	+9.0	13.6	0	+12.0	21.6	0	+20.0	30.5	0	+28.0		
17.72– 19.69	4.4	+1.6	+7.0	7.5	+2.5	+11.6	11.5	+2.5	+15.6	19.5	+2.5	+23.6	26.0	+4.0	+32.5		
	7.0	0	+6.0	11.6	0	+10.0	15.6	0	+14.0	23.6	0	+22.0	32.5	0	+30.0		